But how to realize p-wave pairing?

Minimal model includes:

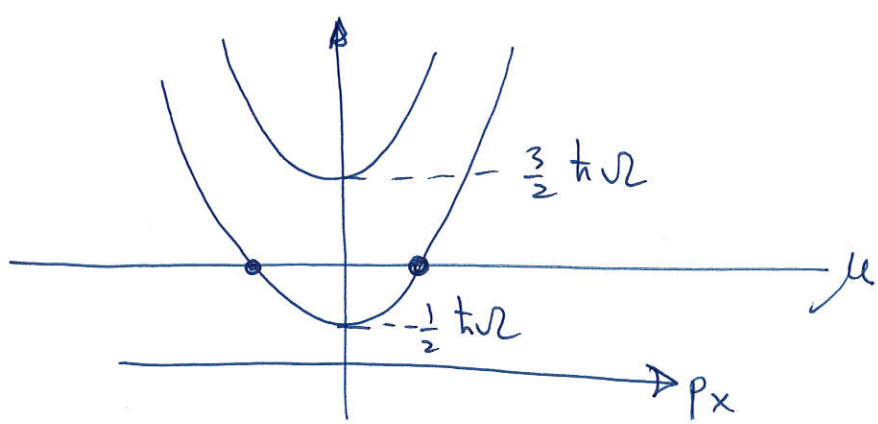
$$\left.\begin{array}{l}\text{Quantum wire}\\ \text{Spin-orbit interaction}\\ \text{Magnetic field}\end{array}\right\} H_{wire}$$

$$\text{Superconducting proximity effect} \Big\} \; H_\Delta = \Delta e^{i\varphi} \psi_\uparrow \psi_\downarrow$$

Wire: $$H_0 = \frac{\vec{p}^{\,2}}{2m} + V(y) = \frac{p_x^2}{2m} + \underbrace{\left(\frac{p_y^2}{2m} + \frac{1}{2} m \Omega^2 y^2\right)}_{\text{transverse quantization}}$$

$$\varepsilon_n^y = \hbar\Omega\left(n + \tfrac{1}{2}\right)$$

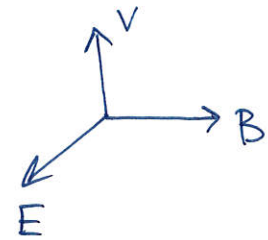
so that $$\varepsilon(p_x, n) = \frac{p_x^2}{2m} + \hbar\Omega\left(n + \tfrac{1}{2}\right)$$



only 1ST transverse subband is occupied.

Spin-orbit interaction of Rashba kind

$$H_{so} = \mu_B \underbrace{\frac{\vec{v}\times\vec{E}}{2c^2}}_{\vec{B} = \gamma \frac{\vec{v}\times\vec{E}}{c^2}} \cdot \vec{\sigma}$$



relativistic effect:

moving electron experiences

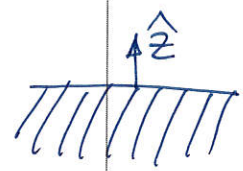
$\vec{B} \sim \vec{v}\times\vec{E}$ ← depends on direction of motion!

But $\vec{E} = -\vec{\nabla}\underbrace{V(\vec{r})}_{\text{potential}}$ → electrons moving in varying potential experience effective magnetic field

$$\Rightarrow H_{so} \sim \underbrace{\vec{\nabla}V}_{\vec{E}} \times \underbrace{\vec{p}}_{m\vec{v}} \cdot \vec{\sigma}$$

Rashba Hamiltonian: at surfaces and interfaces

$$\vec{\nabla}V(\vec{r}) \sim \frac{dV}{dz}\hat{z} \neq 0$$



$$H_R = \alpha_R\, \hat{z}\times\vec{p}\cdot\vec{\sigma} = \alpha_R\, \hat{z}\cdot\vec{p}\times\vec{\sigma}$$

where $\alpha_R \sim \mu_B \frac{dV}{dz}$ and everything else.

$$H_R = \alpha_R\left(p_x\sigma_y - p_y\sigma_x\right)$$

when we quantize motion along y, $\langle p_y\rangle = 0$,

and we are left with 1d spin-orbit Hamiltonian

$$H_{so} = \alpha_R\, \hat{\sigma}^y\, \hat{p}_x$$

$$H_{wire} = \int dx \, (\psi_\uparrow^+, \psi_\downarrow^+) \Big[\frac{\hat{p}_x^2}{2m} - \underset{\substack{\uparrow \\ \text{chem. potential}}}{\mu} + \alpha_R \sigma^y \hat{p}_x - \underbrace{\tfrac{1}{2} g \mu_B B_z}_{\substack{\text{external} \\ \text{Zeeman} \\ \text{field}}} \sigma^z \Big] \begin{pmatrix} \psi_\uparrow \\ \psi_\downarrow \end{pmatrix}$$

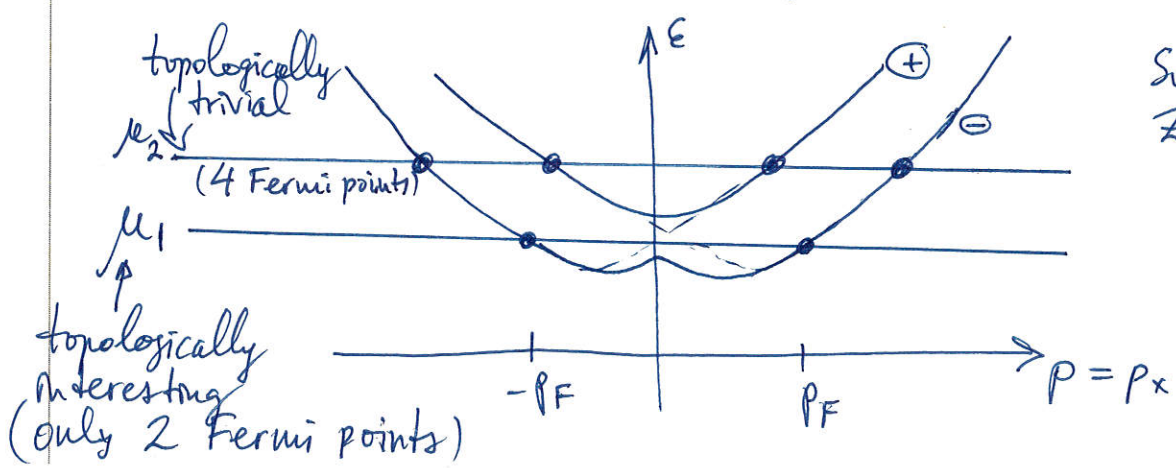
Bandstructure: [denote $\frac{1}{2} g \mu_B B_z \equiv h$]

$$\begin{pmatrix} \frac{p^2}{2m} - \mu - h & -i\alpha_R p \\ i\alpha_R p & \frac{p^2}{2m} - \mu + h \end{pmatrix} \begin{pmatrix} \psi_\uparrow \\ \psi_\downarrow \end{pmatrix} = \varepsilon \begin{pmatrix} \psi_\uparrow \\ \psi_\downarrow \end{pmatrix}$$

$$\det \begin{pmatrix} \frac{p^2}{2m} - \mu - h - \varepsilon & -i\alpha_R p \\ i\alpha_R p & \frac{p^2}{2m} - \mu + h - \varepsilon \end{pmatrix} = 0$$

$$\left(\frac{p^2}{2m} - \mu - \varepsilon \right)^2 - h^2 - (\alpha_R p)^2 = 0$$

$$\varepsilon_\pm = \frac{p^2}{2m} - \mu \pm \sqrt{(\alpha_R p)^2 + h^2}$$



Subbands are Zeeman-split at $p = 0$.

Note: combined effect of Rashba spin-orbit $\alpha_R \sigma^y p$ and Zeeman field $h\sigma^z$ is to break spin-rotational symmetry completely: spin is slaved to momentum

eigenstates $\psi_\pm = e^{i\frac{px}{\hbar}} \chi_\pm$

$$\chi_- = \begin{pmatrix} \cos\left(\frac{1}{2}\delta_k\right) \\ -\sin\left(\frac{1}{2}\delta_k\right) \end{pmatrix} \sim \begin{pmatrix} 1 \\ -\frac{1}{2}\delta_k \end{pmatrix} \quad \text{mostly} \uparrow$$

$$\chi_+ = \begin{pmatrix} \sin\left(\frac{1}{2}\delta_k\right) \\ \cos\left(\frac{1}{2}\delta_k\right) \end{pmatrix} \sim \begin{pmatrix} \frac{1}{2}\delta_k \\ 1 \end{pmatrix} \quad \text{mostly} \downarrow$$

$$\Uparrow$$

$$\delta_k = \arctan\left(\frac{2\alpha_R p}{g\mu_B B_z}\right) \approx \frac{\alpha_R p}{h} \text{ for } \boxed{\alpha_R p \ll h.}$$

Let $\mu = \mu_1$ so that only $(-)$ subband is populated.
Then low-energy excitations with $p \approx \pm p_F$

$$\frac{p_F^2}{2m} - h = \mu_1 \quad \text{in the limit} \quad \boxed{\alpha_R p_F \ll h}$$

which we consider from now on.

~~Then eigenvalue problem is:~~

$$\cancel{\begin{pmatrix} \frac{p^2 - p_F^2}{2m} & -i\alpha_R p \\ i\alpha_R p & \underbrace{\frac{p^2}{2m} + h - \left(\frac{p_F^2}{2m} - h\right)}_{\approx 2h} \end{pmatrix} \begin{pmatrix} \psi_1 \\ \psi_2 \end{pmatrix} = \varepsilon \begin{pmatrix} \psi_1 \\ \psi_2 \end{pmatrix}}$$

Then ψ_- (occupied band) $= e^{ipx}\begin{pmatrix} 1 \\ -\frac{\delta_k}{2} \end{pmatrix} = e^{ipx}\begin{pmatrix} 1 \\ -\frac{\alpha_R p}{2h} \end{pmatrix} = \underset{e^{ipx}\begin{pmatrix}1\\0\end{pmatrix}}{\psi_\uparrow} - \frac{\alpha_R p}{2h} \underset{e^{ipx}\begin{pmatrix}0\\1\end{pmatrix}}{\psi_\downarrow}$

$$\psi_+ \text{ (unoccupied)} = e^{ipx}\begin{pmatrix} \frac{\alpha_R p}{2h} \\ 1 \end{pmatrix} = \frac{\alpha_R p}{2h}\psi_\uparrow + \psi_\downarrow$$

<u>Project</u> Hamiltonian to the lowest occupied subband now: that is, set $\psi_+ = 0$
(no excitations in $\oplus$ subband)

Such projection means $\psi_\downarrow = -\frac{\alpha_R p}{2h}\psi_\uparrow$

Then occupied subband wavefunction

$$\psi_- = \psi_\uparrow - \frac{\alpha_R p}{2h}\left(-\frac{\alpha_R p}{2h}\psi_\uparrow\right)$$

$$= \left(1 + \left(\frac{\alpha_R p}{2h}\right)^2\right)\psi_\uparrow \approx \psi_\uparrow$$

(since we are doing pert. theory in α_R) but everything can be done more accurately for sure.

This projection means that $\psi_\downarrow$ is not an independent degree of freedom as far as excitations in subband $\ominus$ are concerned.

$$\Rightarrow H_{wire} \to \tilde{H}_{wire} = \int dx\ \psi_\uparrow^+ \left(\frac{\hat{p}^2 - p_F^2}{2m}\right)\psi_\uparrow$$

describes spin-polarized electrons.

($p_F = p_F(\mu, h)$ of course, as seen above)

Now observe that superconducting H_Δ (which is treated as weak perturbation to everything else) becomes $H_\Delta = \int dx\ \Delta e^{i\varphi}\psi_\uparrow\psi_\downarrow \xrightarrow[\text{projection}]{} \int dx\ \Delta e^{i\varphi}\psi_\uparrow\left(\frac{-\alpha_R\hat{p}}{2h}\right)\psi_\uparrow$

$$H_\Delta = \int dx \; \Delta e^{i\varphi} \psi_\uparrow \left(\frac{-\alpha_R \hat{p}}{2\hbar} \right) \psi_\uparrow$$

this is exactly a p-wave pairing term we have been looking for.

$$H_\Delta = \int dx \; \frac{i\alpha_R \hbar}{2\hbar} \Delta e^{i\varphi} \left(\psi_\uparrow \partial_x \psi_\uparrow \right)$$

Thus $H_{wire} + H_\Delta \longmapsto H_{p\text{-wave}}$

$$H_{p\text{-wave}} = \int dx \left\{ \psi^\dagger \left(-\frac{\hbar^2}{2m} \partial_x^2 - \mu_{eff} \right) \psi + \Delta e^{i\varphi} \frac{i\alpha_R \hbar}{g\mu_B B_z} \psi \partial_x \psi \right\}$$

Refs: $\begin{cases} \text{Lutchyn, Sau, Das Sarma PRL } \underline{105} \text{ 077001 (2010)} \\ \text{Oreg, Refael, von Oppen PRL } \underline{105} \text{ 077002 (2010)} \end{cases}$

also, Alicea, Oreg, Refael, von Oppen, Fisher Nature Physics 7 412 (2011) discuss non-Abelian statistics of emergent Majorana end modes and their manipulations.

Curious aside: Sau, Gangadharaiah, Starykh considered $\mu = \mu_2$ (trivial topology case) PRB 78 054436 (2008). ☺

$H_{p\text{-wave}}$ is continuous version of Kitaev's [unpaired Majorana fermions in quantum wire, arxiv cond-mat/0010440 (2000)]

H in page 179 $\Rightarrow$ must support Majorana modes at the ends of the open wire.

Even better, these appear at the ~~[illegible]~~ boundaries between $\mu = \mu_1$ (topological) and $\mu = \mu_2$ (non-topological) segments of the quantum wire.

Further review of this all in

M. Leijnse and K. Flensberg, arxiv: 1206.1736v2 (2012)
"Introduction to topological superconductivity and Majorana fermions."